

Agent-based holonic design and operations environment for distributed manufacturing

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Abstract

This paper describes a practical system framework for holonic design and operations in a distributed manufacturing environment using multi-agent systems (MASs) and simulation techniques. Three components are proposed and developed in this system: holonic control, virtual simulation, and human/system integration. With the common system goal as the objective, the holonic control, established upon the JADETM platform in a distributed method, acts as the brain of the entire system. The virtual simulation simulates actions and operations of the physical devices under the holonic control. While the human/system integration facilitates human experts' interaction with the system to choose or override the system's holonic solution. A conceptual material transport system is developed to demonstrate and validate the proposed idea. These key attributes of holonic manufacturing system (HMS), such as robustness when faced with perturbation, autonomous, cooperation and coordination were tested and validated in this open and distributed environment. It shows that the proposed system can satisfy the agility, distribution, and robustness requirements.

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1. Introduction

The increasingly competitive global market makes it difficult for centralized control to cope with different information, expertise and decision-making in the emerging distributed manufacturing paradigm. Effective and efficient cooperation among autonomous entities in an organization is becoming ever important for its success in the competitive environment. In the new paradigm, technologies alone are not sufficient for superior system performance, but have to be integrated with organizational dependencies to fully

realize their benefits [7]. Hence, more and more researchers apply socio-technical methods in order to be able to implement evolution in manufacturing control systems [24,21]. holonic manufacturing system (HMS) [9], as a new paradigm to organize manufacturing activities and meet the agile, scalable, robust and fault-tolerant requirements, overcomes many difficulties faced by existing conventional, rigid systems. Although a number of researchers realize the importance of this concept and engaged in developing holonic technologies for manufacturing systems, few solutions exist for real industrial applications. It is a challenge to build such a system able to provide an effective and efficient way to facilitate holonic design and simulate holonic operations process so as to make the techniques ready for an industrial application.

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As one of the important research topics in distributed intelligence, research in multi-agent system (MAS) has played a key role in the development of HMS. An MAS can be considered as embedding a “general software technology that was motivated by fundamental research questions” [4]. Agent and holon share many common attributes such as autonomous, cooperative and open. Various cooperation, communication, and organizational techniques from MAS literature can be used to implement autonomous, cooperative and recursive agents (i.e. ‘holons’) [3]. The research of MAS provides fundamental technologies for HMS development.

In this paper, multi-agent technology is adopted to realize the holonic control system. Section 2 describes the background of the research activities and analyzes the requirements for the proposed holonic design and operations environment. Section 3 presents the structure of holonic system based on multi-agent technology. Section 4 outlines the system architecture and highlights the three major components, namely, holonic control system, virtual simulation environment and human system integration. Section 5 introduces a conceptual material transport system to test and validate the proposed idea as the first R&D step.

2. Background

HMS research community is organized around the international HMS consortium, one of the projects under the intelligent manufacturing systems (IMS) program [10]. The HMS project is defined as “an international industrially-driven project to address systemization, standardization, research, pre-competitive development, deployment and support of HMS architecture and technologies for open, distributed, intelligent, autonomous and cooperating system through global partnership” [8]. A number of significant results were achieved through the research and development activities conducted in this consortium that covers various topics, such as system architecture and engineering, planning and scheduling, control and holonic man–machine system and emulation, etc. Van Brussel et al. introduced a holonic system architecture, called PROSA [23] that includes three types of basic holons: *order* holons, *product* holons and *resource* holons; Langer proposed a methodology and architecture for

holonic multi-cell control system [13]; Liu et al. presented the architecture and coordination of a holonic automated guided vehicle system [15]; McFarlane and Bussmann developed a holonic production planning and control system [16], and Kotak et al. adopted a virtual manufacturing environment to implement the holonic shop floor control [12].

Inspired by the distributed artificial intelligence, the nascent field of intelligent agent shares similar objectives and features of HMS, i.e. autonomous, cooperative and scalable. Research on agent technology has played an important role in the development of holonic system. In the past years, researchers engaged in various topics on applying the agent technology in the HMS development: Laws et al. present a viable reference architecture for multi-agent supported holonic manufacturing systems [14]; Glanzer et al. implemented a machine-holon using ZEUS agent framework [6]; Aiyama et al. developed holonic robot system using autonomous distributed agent system [1]; Sousa and Ramos introduced an architecture and negotiation protocol for dynamic scheduling using multi-agent and holonic systems [20]. Besides, several open source platforms for implementing FIPA-compliant system, such as FIPA-OS [18], ZEUS [17], JADE [11] and APRIL [2] further facilitate the holonic agent development.

Among these research issues, human expert’s effective interaction with holonic system attracted some researchers’ attention [12]. However, in review the research activities in this area, few published works can be found on this topic. In this paper, we propose a holonic design and operations environment which aims at providing an integrated virtual environment under holonic control to simulate the distributed manufacturing activities or supply chain management in various distributed application domains. The holon’s structure, its assembly in a “holarchy”, the major components of this environment and a conceptual prototype system are described in the following sections.

3. Holon and holarchy based on multi-agent systems

A holon is defined by the HMS Consortium as “an autonomous and cooperative building block of a manufacturing system for transforming, transporting,

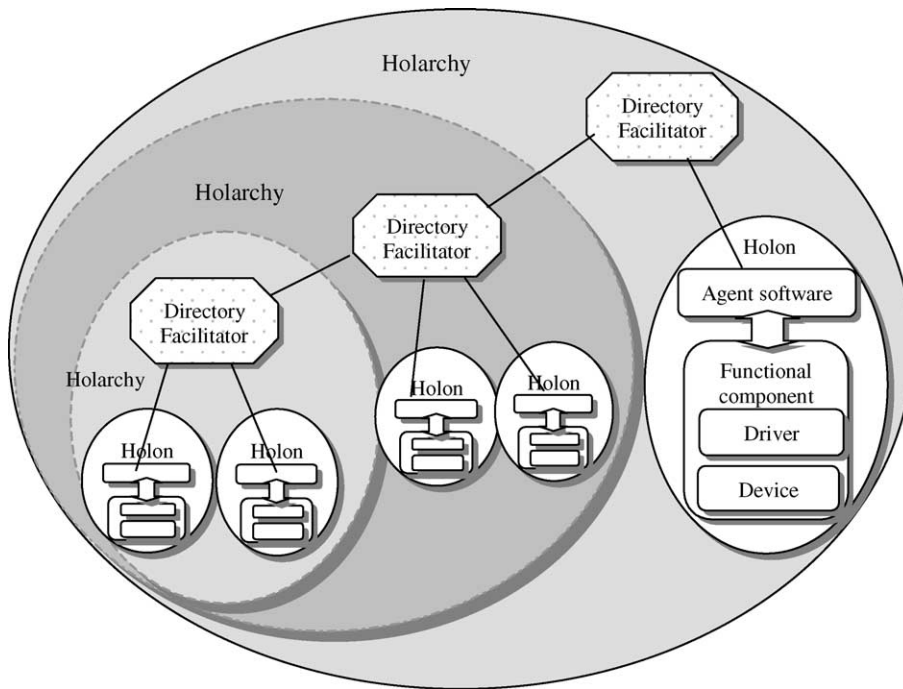


Fig. 1. Agent, holon and holarchy.

storing and/or validating information and physical objects”[3]. A holon in a holarchy is characterized by autonomy and cooperation generally consists of a physical component and an informational component. The overall system objective is achieved by holarchy that is composed of basic holons or recursive holons that may also be another holarchy. The communication and coordination are conducted through a concept of cooperation domain (CD) that includes a physical infrastructure for communications and facilities for coordination, data storage and other communication facilities.

Agents, as they are autonomous, cooperative and distributed in nature, are chosen in this research to build the communication and cooperation part of these dynamic and interactive holons. Each individual basic holon has at least two basic parts: functional component that could be pure software or a software entity representing an interfacing hardware (physical device); communication and coordination component that is usually realized through the implementation of agent software. In our research, the functional component in each holon is implemented as a software

entity that can be developed in any program language in any environment and be wrapped by its corresponding agent software, while the directory facilitator in a multi-agent system is equivalent of the CD in HMS. Fig. 1 shows the basic structure of a single holon and holarchy formed by MAS.

The holonic agent in this research is an extended JADETM agent [11], which is fully implemented in the JavaTM language. The JADETM agent is platform independent, therefore, the system can be distributed across different machines with different operating systems. System configuration can also be controlled and changed through a remote GUI, even the agents can be redistributed at run-time and moved from one machine to another as required. The function of CD in HMS is achieved through the directory facilitator (DF) and agent management system (AMS) in the JADETM platform. Fig. 2 illustrates the architecture of the holonic agent implementation.

Physical devices controlled by holons are implemented as software functional components in this paper. These virtual components are driven by the device driver to simulate the operations and activities

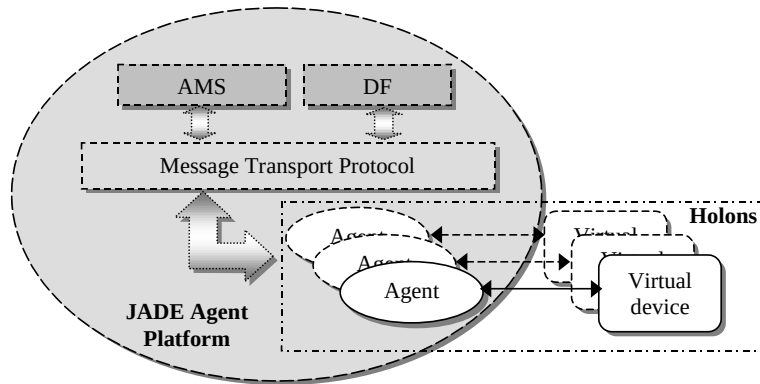


Fig. 2. Architecture of holonic agent implementation.

of each individual physical device in the real world in a virtual environment. Agent, device and the corresponding driver are integrated together to form an individual holon in the system (see Fig. 1). Without losing generality, these components can be implemented in different environments using different languages.

4. Holonic design and operations environment

In order to reach the goal of reliability, flexibility and productivity in a HMS, the realization of the autonomous and cooperative holons in a holonic control system and an effective mechanism for its integration with human decision makers are among the key issues.

Also, prior to the integration of the designed holonic manufacturing system with a physical manufacturing environment, a virtual simulation environment can provide an efficient tool for facilitating system design and enhancing operational performance. Therefore, three major components are considered in the proposed holonic design and operations environment, namely, holonic control system, virtual simulation environment and human system integration. Fig. 3 illustrates the general system structure.

4.1. Holonic control system

Holonic control is usually described as a concept that combines the best features from both hierarchical and heterarchical control. According to van Brussel

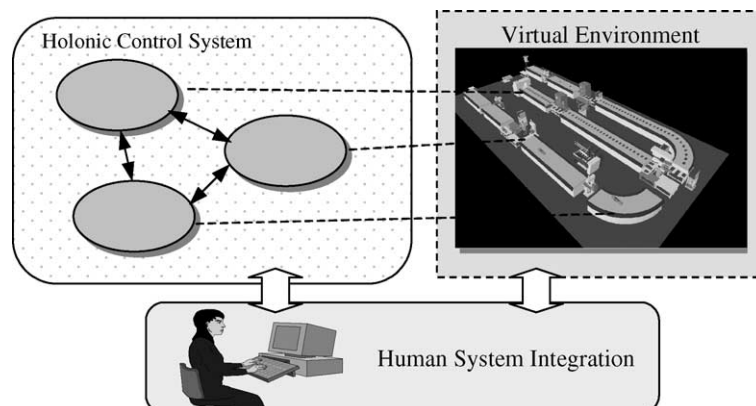


Fig. 3. Holonic design and operations environment.

et al. [22], holonic control differs from traditional control on the following aspects:

- Holons can belong to multiple hierarchies.
- Holons can form temporary hierarchies.
- Holons do not rely on the proper operation of each holon in the hierarchy to get their work done.

These attributes of holons make them suitable for realizing the concurrency in manufacturing engineering and solving the problem of increasing dynamics in manufacturing companies. In the holonic control system, each autonomous entity in the manufacturing system is modeled as an individual holon. Each holon works independently and they can dynamically coordinate and cooperate each other to reach the common system goal.

The holonic control system acts as an ‘engine’ of the entire holonic design and operations environment. As illustrated in Fig. 1, coordination and cooperation among ‘devices’ are achieved through the communication of their wrapping intelligent agents, which are composed of several component modules, namely, a decision-making module, an execution and monitoring module, a communication and cooperation module, and an objective and beliefs database. Fig. 4 illustrates the basic structure of each holon in this control system.

4.1.1. Decision-making

The main function of decision-making module is two-fold: one is to manage existing plans and create a new one when the corresponding tasks included are

ready to be executed; another one is to decide when to execute these plans, i.e. scheduling (used by the execution and monitoring module). Each holonic agent has its own autonomy to make its own decision for planning and scheduling to achieve a common system goal. This module is the ‘brain’ of a holon and controls all activities and information flow.

In the planning and scheduling process the holon uses the knowledge and information about itself and its environment, then cooperates and coordinates with other holons in the system for a specific task. The holons are also intelligent and make decision based on the knowledge acquired; they also have the ability to learn from previous experiences to further enrich their knowledge base.

4.1.2. Execution and monitoring

In our system, this module is used to control and monitor the operational execution of the schedule in corresponding physical device or legacy system that was implemented as virtual entities in the simulation environment. The functional component is a major part of this module that includes a functional control manager, a device driver, a monitor, and a command dispatcher.

The functional control manager is used to manage and control a set of scheduling commands while the command dispatcher dispatches the individual command to the corresponding device. The device driver provides a mechanism to communicate with the physical devices or legacy software systems, while the monitor allows the passive and active monitoring of

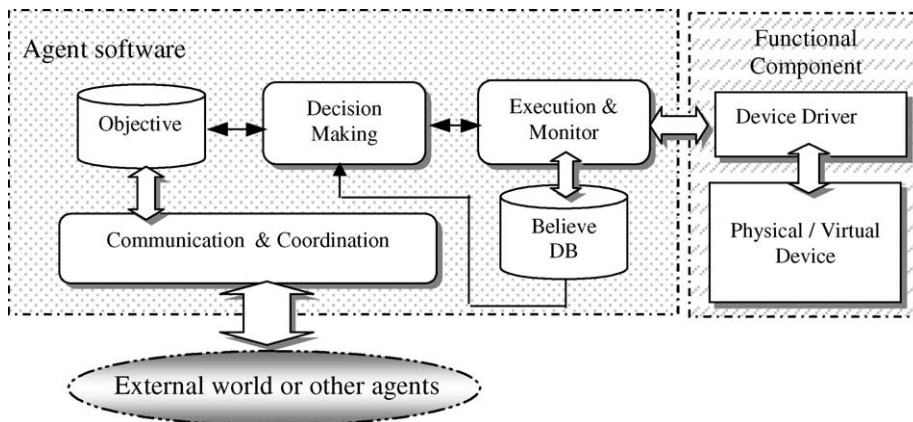


Fig. 4. Structure of holon in the control system.

the environment changes that is used to update the holonic agent's belief database.

4.1.3. Communication and cooperation

The overall system goal is achieved through cooperation and coordination among distributed holons in the system. Communication and negotiation are the way to realize these functions. Three key issues, namely, communication language, negotiation protocol, and ontology are included in this module.

The FIPA agent communication language (ACL) [5] is adopted as the communication language among holons. FIPA-ACL is a part of FIPA efforts dedicated to promoting industry of intelligent agents by openly developing communication specifications supporting interoperability amongst agents and agent-based applications. Communication is the means for negotiation; negotiation protocol sets the stage for the negotiation process and indicates when and how to exchange offers among agents using contract net protocol (CNP) [19]. The format of messages for data translation uses the eXtensible markup language (XML) language. The receiving agent uses an ontology for the standardization of the message's contents to support the interpretation of the content expression. The concept of "Ontology" is commonly understood by the agent community, and is domain dependent. Examples are given in our conceptual prototype system in the following section.

Based on the demands from other holons or other modules within the same holon, the communication and cooperation module uses the predefined TCP/IP protocol to asynchronously communicate with other holonic agents in several modes: one to one, multicast or broadcast.

4.1.4. Objective and belief database

Requests to a holonic agent are transformed through the communication and coordination module into its objective database. The objectives of a holon are stored in its objective database in a prioritized queue. The objective with highest priority will be selected by the decision-making module that consequently constructs a plan and schedule to reach a goal.

The domain knowledge of each holonic agent is stored in its belief database, which provides the support to the decision-making module for the right decisions. Beliefs of each holon about the environment can be

dynamically updated with the execution of commands and changes that occur in the environment.

All of the above modules in an individual holon are implemented as autonomous threads of control to allow concurrent communication, execution, and monitoring. Actions themselves are also executed as separate threads and can run concurrently so that some of the actions, such as request of services to other remote agent are enabled through this actual parallelism.

4.2. Human system integration

A human expert's interaction with an HMS is an essential component in the proposed system. Human system integration component provides an efficient mechanism to support the human expert in the decision-making process, system configuration and operations. The role of the human system integration in this research is two-fold, the first is to initialize the system configuration and dynamically control the system change in the simulation environment to test some key attributes of a HMS; the second role is to provide a mechanism to enable the human expert to be involved in the decision making process when necessary. The human/system integration is made through the implementation of human system interface within holons through which the human expert can interact with other system components, input his decisions and participate in the negotiation process.

The integration of humans with the holonic system is through event commands. Therefore, in addition to the major components of a general holon as defined in the previous section, an additional component named human system interface can be added through which a human expert is able to communicate with the system. Fig. 5 illustrates the basic structure of the human/system integration in a holon.

The learning ability of holons is used to capture the tacit knowledge from the decision making process made by human experts, which will facilitate the future decision making process and enhance the decision making ability of the corresponding human holons.

The implemented human system integration is proven to be efficient and effective for achieving optimal system performance. The example in Section 5 is a conceptual prototype system we developed in this research, in which the human/system integration is achieved through the graphical user interface. Some

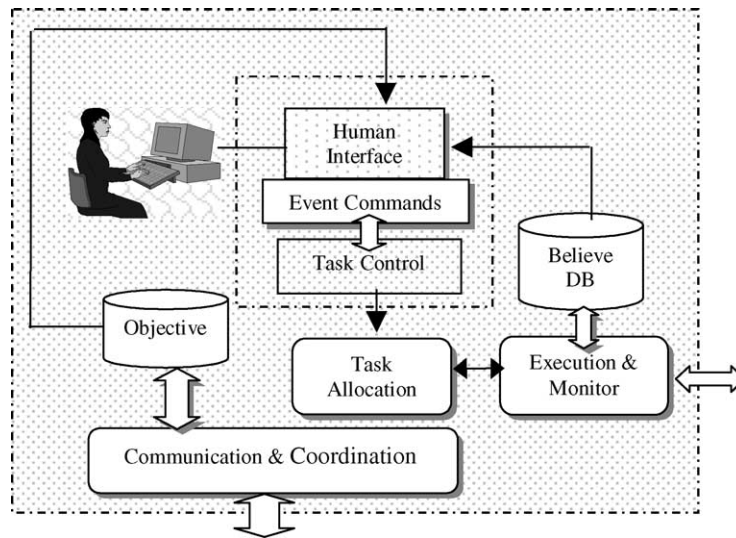


Fig. 5. Human system integration.

key system attributes such as system robustness during perturbation is tested and holonic cooperation and coordination is also validated. Another application of human/system integration was implemented in an industrial project for operational scheduling in the shop floor level. Here, the human/system integration module provided to the human expert an easy method to interactively create the operational schedule by using the human expert's tacit knowledge and integrating it with a 3D virtual simulation environment.

4.3. Virtual environment

The basic approach to demonstrate and validate the proposed concept and design is to use a virtual simulation environment. Driven by the external holonic control system, either a 2D simulation environment or a 3D discrete event simulation model can be adopted to emulate system operations, predict system performance and validate the proposed holonic control system design. A 2D simulation developed in Delphi is adopted in our conceptual prototype system which provides enough flexibility for the human user to 'disturb' the system while it is running so that some key attributes of HMS can be tested, such as robustness in the face of perturbation. A 3D simulation model using QUEST was developed in another industrial project which focused on operational scheduling.

It was integrated with our holonic manufacturing system.

The operational activities in the 2D/3D simulator are controlled by the holonic control system, which is implemented based on the distributed multi-agent technology previously described. Holonic agents run on one or more computers in a local area network with the proprietary TCP/IP based protocol. A holon acts as the controller of the individual physical equipment, while the controller is linked to the corresponding device driver and virtual device implemented in the 2D/3D system simulator. Drivers in a 2D/3D simulator are implemented as individual 'logics' that are used to simulate the system operations. Fig. 6 shows the links between the simulator and holonic control.

The virtual environment can be used to simulate various manufacturing processes for different purposes. In this research, the virtual environment is used to assist the holonic design and system layout, emulate the system operations that are controlled by the external holonic control system and assist in evaluating the holonic system performance.

5. Example

In order to validate the proposed concept, structure and techniques, a rapid conceptual prototype of material

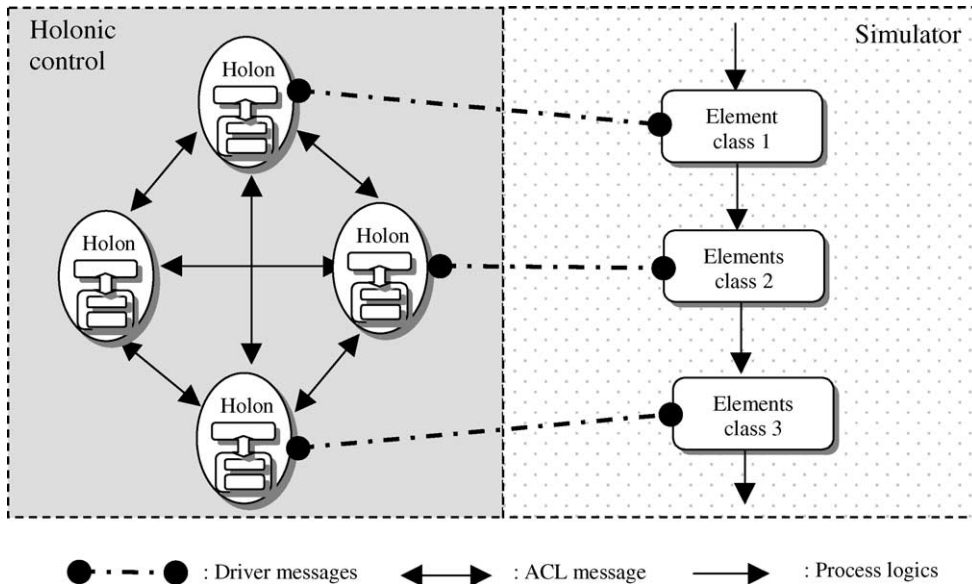


Fig. 6. Virtual environment driven by holonic control.

transport system was developed. The transportation problem in this system was chosen and emphasized because of the diversity of its layout designs and flexibility of changes so that the holonic cooperation and coordination can be simulated and holonic system performance can be easily evaluated while experiencing perturbation and dynamic change.

5.1. System requirements

The general objective of a material transport system is to transport work pieces (parts, raw material, etc.) to their destinations with the lowest cost, time or distance. In this conceptual prototype system, a 'part' entity (representing workpieces) with certain attributes can be generated by the 'source' entity (representing inventory) and be transported by 'conveyor' entity to its destination 'sink' entity through the selected optimal transportation path. The system should have the ability to adapt to the dynamic changes of system layout; any failure of a single entity in the transport system should not make the entire system 'breakdown'. In other word, each entity in the transport system is autonomous and cooperative, and their implementations are independent from the entire system configuration. Therefore, key attributes of the holonic manufacturing system can be tested, namely, autonomy, cooperation and cooperation,

open and robustness. The following standards are defined for evaluating the system performance:

- *Open*: An arbitrary number of system entities can be added and deleted even when the system is running.
- *Robustness*: Dynamic change of system configuration, enabling or disabling of any individual system entities should not affect the system performance.
- *Autonomy*: Each individual system entity is autonomous and makes its own decision independently; its implementation is independent from the entire system.
- *Cooperation and coordination*: The overall system performance is achieved through the cooperation and coordination among system entities.

5.2. System design

In order to meet the above requirements, five different kinds of entities were designed and implemented in the conceptual prototype material transport system: source, part, conveyor, sink and decision-points. The number of each instance of entity appearing in the system is arbitrary and the layout can be dynamically changed so that the perturbation and open attributes of a HMS can be achieved. Fig. 7 illustrates the relationship of the different components.

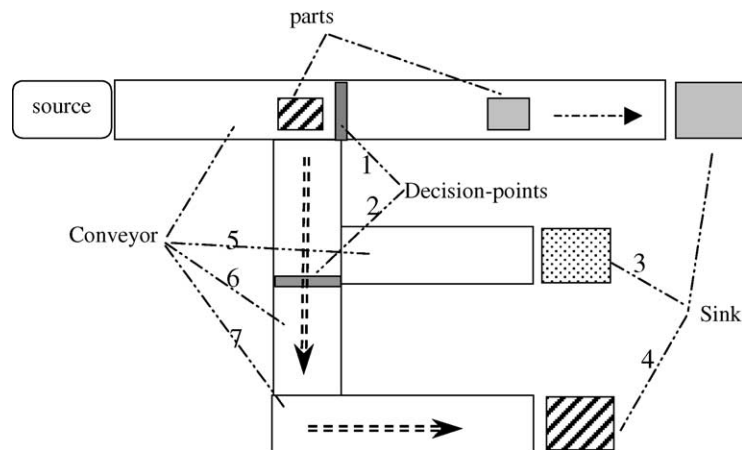


Fig. 7. Relationship of different entities in the conceptual prototype system.

The “part” component was generated from a source entity with certain attributes and can be transported by the conveyor to a sink destination that has the same attributes. The “decision point” entity always sits on the intersection where two or more choices of possible output paths exist. Through cooperation and coordination among different system entities, an optimal path with lower “cost” (which in this research is simplified as the shortest distance) will be chosen to transport the part to its destination.

5.3. Holonic control system implementation

The conceptual material transport system is implemented as an agent-based holonic system in the holonic design and operations environment. Each entity controller is implemented as an agent that integrates with the virtual entities in the simulation environment to form a holon. Each holon is autonomous and has the responsibility of maintaining its local knowledge and keeping its neighbor holons informed once something changed. None of the individual holons has the knowledge about the whole system.

The system layout design is configured by a human system interface through which human users can feel free to add or delete any type of system entities. Once a system entity is added to the system, a new holonic agent is automatically created and its relationship with other holons (i.e. inputs and outputs relations) will be

stored into its belief database. Through the dynamic information exchange among holonic agents, the transport network of the entire system is distributively stored in each individual holon and any change of the system layout can be detected and stored. Through the integration of holonic agent control system with the simulation environment, a holonic design and operations environment is formed.

Holonic operations, i.e. moving parts from source to their destinations, are achieved through the cooperation and coordination among related holonic agents. A human system interface is used to let user interactively change the system configuration and communicate with other holonic agents to adapt to the changing environment. As previously described, the JADETM system is used as the developing platform and the eXtensible markup language (XML) is embedded in FIPA standard ACL message for data communication among agents. The following sections briefly explain this implementation.

5.3.1. Communication and cooperation

Each individual task is performed through the communication and cooperation among holons. Take the part generation as an example, a part is generated upon the request of certain destination (sink) entity, which is implemented as the REQUEST performative as in FIPA ACL standard. For example, suppose sink1 requires a part from source2, the process and message will be as:

```
(
  :REQUEST
  :sender sink1
  :receiver source2
  :content (<newpart>XMLMessage</new-
part>)
  :language XML
  :ontology ML_HDOES
  :protocol fipa-request
)
```

where the XMLMessage above is like:

```
<part No = 012002" attribute = "gray" sink =
"sink1"/>
```

Once the source2 receives this message, it will check if the required part is available, if there is no such items, a message with “REFUSE” performative will be replied to the sender; otherwise, a message with “AGREE” performative will be replied, then another new message with “INFORM” performative and this part information will be sent out to the next corresponding conveyor holon. At the same time, a new mobile part holon with the number 012002 and attribute ‘gray’ will be loaded and be conveyed by the conveyor. This process is illustrated in Fig. 8.

As illustrated above, a FIPA-ACL message contains a set of one or more message elements. Precisely which elements are needed for effective agent communication will vary according to the situation. However, the *performative* element is mandatory in all ACL messages and *sender*, *receiver* and *content* elements are usually expected in most ACL messages.

5.3.2. Decision making and execution

Decision-making is made independently by individual autonomous holons through cooperation with other holons. Take the choice of optimal path for the part as an example. Once the part holon is generated by the source holon, the part holon will inform the conveyor holon about its arrival and the conveyor holon will start to move the part forward until it arrives at a decision-making holon or reaches the destination sink holon. As the decision-making holon always sits on the crossing intersections in the system, it has to make a decision as to which path the arriving part will be switched. Each decision holon maintains its own knowledge about the destination sinks that are reachable by each of its alternative output conveyors together with the transport cost factor. Once a part arrives, the decision holon will search its knowledge base to find out an optimal output path and negotiate with the corresponding conveyor holon. Once an agreement is reached, the part will be switched out to the crossing conveyor or continue on its current path.

Once the part reaches its sink destination, the part will be ‘consumed’ by the sink destination and the corresponding part holon will be destroyed.

5.3.3. Open, robustness and adapt to perturbation

According to the requirements, system running performance should not be affected by perturbations, such as failure of a system component or change of system configuration. In order to meet these requirements, the local belief knowledge base maintained by each individual holon and the cooperation among holons through backward propagation message passing are implemented in this system. For example, if

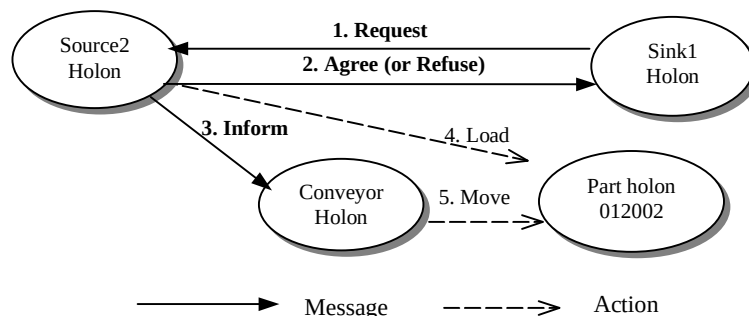


Fig. 8. Communication among holons.

sink3 is deleted from the system, an INFORM FIPA ACL message will be backward propagated to its preceding holons and the sink3 will be deleted from the reachable list stored in the belief knowledge base in these holons. This process continues until no preceding holon exists so that the path to this destination will become unreachable in the system. In the same way, if any holon changes its position, the same back propagation process will propagate throughout the system, so that each individual related belief-databases will be updated.

Back propagated message-passing keeps each individual holon independent and related to each other. Therefore, the system is open, the user can feel free to add or deleted any system entities to change the configuration. Furthermore, if any single holon failed, the only reaction of other holons is to update its own belief database and its decision making logic is not affected. Therefore, the system is robust and stable when facing any unexpected perturbations.

5.3.4. Ontology

Ontology represents a common structure so that holonic agents can use the same semantics of terms in the message for communication and exchange data information. In this research, the XML is adopted for a set of ontology specification. ML_HDOES is defined for the test case of the conceptual prototype material transport system in the holonic design and operations environment.

Some of the XML descriptions for ML_HDOES, such as “part”, are already shown above. All other system layout information distributively stored in each knowledge base of individual holons is also implemented in the same way. For example, the decision-making point D2 in Fig. 7 can be described as:

```
<component type = “decisionMaker” ID =
“D2”>
  <from = “D1”/>
  <to = “S3” next = “C5” cost = “10”/>
  <to = “S4” next = “C6” cost = “12”/>
</component>
```

and sink3 can be described as:

```
<component type = “Sink” ID = “S3”>
  <from = “D2”/>
</component>
```

If another sink named S8 and a conveyor C9 is added to the system and if S8 is reachable from D2 through C9 with a cost value 20, then information

```
<to = “S8” next = “C9” cost = “20”/>
```

will be passed and be added to the D2’s knowledge base, and the system will be dynamically updated.

5.4. Test and result

The conceptual material transport prototype system is developed and implemented as the first step to test and validate the proposed idea and technologies adopted. A 2D GUI interface shown in Fig. 9 illustrates one of the arbitrarily user designed system layouts, where the user can feel free to change the system layout or add/deleted any of the system entities even while the system is running. Through a number of tests and evaluations, the following conclusions were made:

- *Open and flexible configuration*: Users can arbitrarily design the system layout without any change to the holonic agent software implementation, therefore, the system is open and flexible.
- *Robustness*: Users can dynamically change the system configuration, add or remove any system entity without an entire system break down.
- *Autonomous, cooperation and coordination*: Each holon runs autonomously and makes decisions independently; they dynamically cooperate and coordinate with each other, and the optimal solution can be achieved through individual decisions based on the current local and global situation.
- *Holarchy*: Each entity in the material transport system is implemented as a holon, and the function of cooperation domain is achieved through the implementation of JADETM platform. Therefore, the system can also be taken as a recursive holon when it is integrated with other manufacturing subsystems, such as a group of manufacturing cells. In this case a holarchy will be formed as in Fig. 1.

As the first step of developing and implementing the holonic design and operations system, the concept and technologies were validated, the original objective was achieved, thus, paving the way for further development and application.

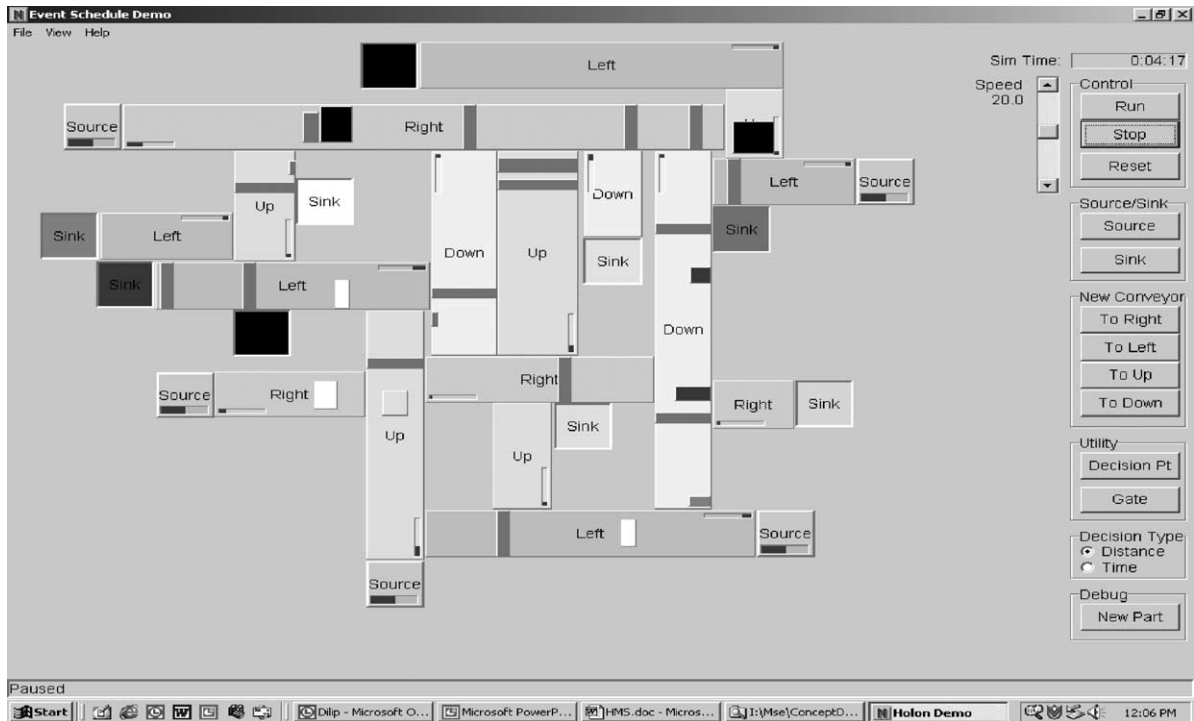


Fig. 9. Example of a conceptual holonic material handing system.

6. Conclusions

This paper proposes a concept of holonic design and operations environment and describes a developed prototype system—a holonic material transport system using this concept and existing technologies. Three major components are proposed and developed in this environment, namely, holonic control system, virtual simulation environment and human system integration. The proposed concept was validated and these key attributes of HMS, such as robustness when faced with perturbation, autonomous, cooperation and coordination were tested and validated in this open and distributed environment.

The holons in a holarchy are realized using intelligent agent technology and implemented based on JADETM FIPA-compliant open sources. FIPA-ACL is adopted for the communication among holonic agents and XML is embedded for message expression. A 2D virtual simulation environment is integrated with the holonic control system, while a 3D discrete event simulation model using QUESTTM is also being integrated with a holonic manufacturing system in our

industrial projects. In the conceptual prototype system, human system integration is mainly used to provide the human user a means to design the system, ‘disturb’ the system and dynamically communicate with the holonic control system to change system environment.

As one of the HMS phase 2 tasks, our present research work in holonic design and operations environment is still in the preliminary stage. Based on the control architecture and holonic technologies used in this conceptual prototype system, our on-going development will focus on holonic application in distributed energy generation and distribution in energy industry. Artificial Intelligence, such as neural-fuzzy technology will also be implemented to facilitate holonic decision-making processes to enhance its learning ability.

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